

Biological Psychology (PSYC 220), Section 003

Tue/Thur, 2:00 – 3:15 PM, Dey Hall-Rm 0305

Course site: <https://uncch.instructure.com/courses/81433>

E-mail: PSYC_DrC@unc.edu

Department of Psychology and Neuroscience

Instructor: Dan Christoffel, Ph.D.

Office: Davie Hall, Room 122

Zoom ID: <https://unc.zoom.us/my/drchristoffel>

Office Hours: Monday 4:00 – 5:00 PM
(or by appointment)

The instructor has the right to make ANY changes to the syllabus when unforeseen circumstances occur. These will be announced ASAP so students can adjust accordingly.

Course Description

The central nervous system is responsible for complex behaviors and mental processes. This course is designed to help students arrive at informed answers to questions about how the nervous system is constructed and assembled (cells and structures), how the brain detects and incorporates sensory information (vision, hearing, taste, smell), and how the brain regulates other basic functions (sleep, hunger, emotion). *This course will have a strong focus on the anatomical, physiological and molecular aspects of the brain that govern its functioning in these systems and behaviors.* In addition to the assigned textbook students will get exposed to primary literature during in-class activities and learn how we study the nervous system. Besides providing an excellent foundation for further study in the field this course is designed to support active student engagement.

Learning Objective

You will gain a detailed understanding of the divisions, structures, and functions of the nervous system. You will be introduced to reading primary literature and learn how to extract information as well as discuss its content via group in-class activities. Additionally, you will learn to appreciate how approaching psychology from a biological perspective has given invaluable insight into several questions. For example: **How are we able to feel, hear and see? How is our body able to regulate basic functions, such as sleep, thirst, and hunger?**

By the end of this course, students should have a good core set of knowledge and skills on how to begin answering these questions, where to find the answers, as well as how to think critically about the answers given.

Specifically, by participating in this course you can learn the following:

- Define & explain terms to describe nervous system organization and function
- Examine key steps of specific biological processes
- Examine psychological concepts from a biological perspective
- Work with primary literature, and learn how to extract information, as well as discuss it
- Overall, this course will provide an excellent opportunity to understand the basic principles of biopsychology for further study in the field

Office Hours and E-mail

Office hours provide an opportunity for you to meet with either me individually (or in small groups) outside of class. This time can be used to receive help with or dive deeper into course material, ask for feedback or additional guidance on coursework, as well as discuss broader career development questions. **Only use the above course e-mail**, which will be primarily answered during the hours of 9am – 5pm M-F.

Required Course Material

Text: Kalat, James W., Biological Psychology (12th ed or up) Belmont, CA: Thomson Wadsworth. Available in Student Stores. You do NOT need to purchase MindTap® Psychology. The book is also available as an eBook via [Cengage](#).

Library Reserve

The textbook for this course is available on reserve at the Davis Library. I am committed to ensuring that this course is financially accessible to all students. Please reach out if you have concerns about the affordability of materials, and we will figure something out. Note: Chapters 1 and 2 (13th ed) are digitized and available via “Course Reserves” tab of our course’s Canvas page.

Additional Reading: For some topics we will also dive into primary literature (journal articles) which will be distributed on Canvas.

Lecture Notes: PowerPoint presentations (converted into pdf) will be posted prior to class on Canvas and are equivalent to handouts. It is recommended that you have the lecture notes available while reading the assigned book chapter(s) as it will help you to focus your reading on the topics discussed in class. Readings should be done prior to class as corresponding quizzes will have to be completed before class. Lectures will focus on selected concepts discussed in the assigned chapter(s) and in-class activities will be used to help build on your knowledge. Further, the lecture notes should be used as a study guide for the exams.

Canvas Site (onyen login): Canvas (<https://Canvas.unc.edu>) will contain course information, access and download lecture notes and primary literature, take quizzes, submit your final paper, and check your grades. Please check your Canvas profile to ensure the correct email address is listed. This is critical and must be accurate in order for you to receive notifications of uploaded documents, such as readings and assignments, as well as class emails and announcements!

PollEverywhere (you will need to login to be able to respond): You must be a registered user to be able to respond to PollEverywhere. If you do not have a PollEverywhere account please go to: <https://poll.unc.edu> and create your student account by clicking on “Student Account Registration” on the top of the website.

Grading Components

Assessments*	Percentage
Exam 1	12%
Exam 2	12%
Exam 3	12%
Final Exam	16%
Papers (4 out of 4)	12%
Quizzes (16, lowest 2 are dropped)	14%
Attendance & In-Class Activities	10%
Underrepresented Scientist Spotlight	7%
Peer Evaluations	5%
TOTAL	100%

*Please note: Due to time constraints, any quiz or paper issues must be resolved within 2 weeks of the assignment. Requests later than 2 weeks will not be considered.

Letter Grades

Grade	Percentage	Grade	Percentage	Grade	Percentage
A	92.5-100%	B-	79.5-82.4%	D+	66.5-69.4%
A-	89.5-92.4%	C+	76.5-79.4%	D	59.5-66.4%
B+	86.5-89.4%	C	72.5-76.4%	F	<59.4%
B	82.5-86.4%	C-	69.5-72.4%		

Honor Code: A condition of joining the Carolina community, Carolina students pledge “not to lie, cheat, or steal” and to hold themselves, as members of the Carolina community, to a high standard of academic and non-academic conduct while both on and off Carolina’s campus. This commitment to academic integrity, ethical behavior, personal responsibility and civil discourse exemplifies the “Carolina Way,” and this commitment is codified in both the [University’s Honor Code](#) and in other University student conduct-related policies.

Use of generative AI tools of any kind is not permitted in this course. Any use of these tools will be considered an instance of academic dishonesty and will be referred to the Honor System.

Diversity & Inclusion Statement

I value the perspectives of individuals from all backgrounds reflecting the diversity of our students. I broadly define diversity to include race, gender identity, national origin, ethnicity, religion, social class, age, sexual orientation, political background, and physical and learning ability. I strive to make this classroom an inclusive space for all students. Please let me know if there is anything I can do to improve. I appreciate any suggestions.

In an ideal world, the practice of science would be completely objective. However, while the scientific process is designed to eliminate bias and subjectivity, scientists are still human beings and as psychology and neuroscience have taught subject to a variety of biases. Furthermore, scientific literature is NOT simply a reporting of facts but involves interpretation and as such it is of the upmost important to think critically about the data, reported results and hypothesis in general. I acknowledge that it is possible that there may be both overt and covert biases in the material due to the lens with which it was written, even though the material is primarily of a scientific nature.

Class Guidelines & Learning Environment

I intend to create a learning environment for students that supports a diversity of thoughts, perspectives and experiences, and honors all identities (including race, gender, class, sexuality, religion, ability, etc.). To help accomplish this:

- If you have a name and/or set of pronouns that differ from those that appear in your official UNC records, please let me know!
- If you feel like your performance in the class is being impacted by experiences in or outside of class, please don’t hesitate to come and talk to me. I (like many people) am still in the process of learning about diverse perspectives and identities.
- As a participant in course discussions, please honor the diversity of your classmates

I expect that everyone will respect our learning environment and each other. If you ever feel that your right to learn and participate in this course has been violated by myself or any of your peers, please do not hesitate to contact me.

As your instructor I affirm my commitment to:

- respect the dignity and essential worth of all individuals
- promote a culture of respect throughout the university community
- reject bigotry, discrimination, violence, or intimidation of any kind
- practice personal and academic integrity and expect it from others

Policies & Procedures

In-Class Exams (3 Class Exams, each 12% of your course grade)

Exam Policies: The three in-class exam formats will range from multiple choice, opened ended, matching, and fill in the blank. You will have 75 min for Exams 1-3. Exams will be closed book.

Make-Up Exams: Make up exams will only be administered for [university-approved absences](#). It is your responsibility to contact me for setting up a make-up exam. **Exams must be made up within one week of the test date.** If you miss an exam without a valid excuse, you will receive a zero.

ARS: If you are eligible for exam accommodations through [Accessibility Resources & Service \(ARS\)](#), be sure to plan with them at least one week in advance of each exam.

Reviewing Exams: You will only be able to review each exam for 1 week following grading. Please don't delay in reviewing your exams! Also, during the LDOC we will review all three exams before the final, and they will be made available for review until the Final exam.

Final Exam (One In-Class Final Exam, 16% of your course grade): The final in-class exam will be cumulative and will be closed book. It will only contain multiple choice questions. You will have 3 hours to complete this exam. All students must take the final exam (this is University policy). If you miss the final, you will be given an "AB" – if you have an official exam excuse, you'll need to take the final at an alternate time. [Final Exam Schedule Spring 2025](#)

Papers (12% of your course grade): There will be 4 papers that ask you to think more in depth about specific topics we will be covering in class. **You are responsible for completing ALL 4.** Each paper is worth 100 points, and will be graded based on the following:

- All questions in the prompt are answered fully (30 points)
- The answers are accurate, complete, and based on material from lecture or textbook (50 points)
- The paper is at least 700 words in length (10 points)
- The paper is formatted using full sentences and paragraphs - no bullet points (5 points)
- Spelling and grammar are correct (5 points)

DO NOT IDENTIFY YOURSELF IN YOUR PAPER SUBMISSION. I will grade them in a blinded manner. **You will lose 5% of the total points if your name is on the paper.** Some of these papers will require you to read a journal article, visit a website, etc. Paper due dates appear in the syllabus in **red** and are due at before class. **Papers that are submitted late will lose 5% of the total points for each day past the due date.** If you require a make-up for a paper you must acquire an official excuse and contact me. Papers must be submitted via Canvas. Papers are open book but should be completed independently. It is an honor code violation to work with other students on these papers or to plagiarize information. Use of AI is not permitted.

How to submit.

- Assignments will be posted on Canvas ("Assignments" tab)
- Submit via the GradeScope Link
- Regrade Requests must occur **within 1 week of grade posting**

Quizzes (16, drop lowest 2, 14% of your course grade): Quizzes (multiple choice) will have to be completed online, are open-book, and **are due before class**. Quizzes are based on the textbook. Quiz dates appear in the syllabus in **blue**. If you complete all 16 quizzes your lowest two will be dropped. Extras will not earn extra credit. If you require a make-up for a quiz you must contact me to coordinate a make-up within 1 week prior to quiz.

Attendance & In-Class Activities (10% of your course grade): We will have in-class activities throughout the semester where you will be asked to complete different class activities/exercises (individually, with a partner, in a group). Sometimes you will be submitting these assignments to earn credit (including PollEverywhere questions). You need to answer them correctly to reach full points (points will be taken off if the answer is only partially answered, if it is missing important points, etc.). If you read this sentence and bring it to my attention by email before Exam 1, you will receive 3 extra points towards your final grade. If you have an excused absence, you must coordinate a make-up assignment with me to receive points for the missed class within 1 week.

Underrepresented Scientist Spotlight (7% of your course grade): To further connect our course to our lives, we are going to create spotlights for scientists we admire. This project will specifically focus on scientists that are underrepresented ([NIH Diversity Statement](#)) in science. Please refer to the rubric posted on Canvas for specific instructions.

Peer Evaluations (5% of your course grade): Research shows that students working in small groups often learn more and demonstrate better retention than students taught in other instructional formats. Peer evaluations will be used for the underrepresented scientist spotlight assignment to ensure that all group members are working together and doing their parts. Please refer to the rubric posted on Canvas under “Resources” for specific instructions.

Extra Credit: Extra credit opportunities will be available in exams. Extra credit points will factor into each of the exams.

Study Tips

Carolina can be challenging but everyone can do well if you 1) know how to study and 2) take time to study. The goal is not to study 24-7 but to study efficiently (i.e., study for the least amount of time for the most amount of information). Here are tips on how to do this:

1. Attend each lecture, pay attention and participate.
2. *Keep up with material.*
3. Read the relevant chapters *before* each lecture. Also it will help to download the ppt slides and review it while reading the assigned book chapter(s) and/or journal articles as it will help you to focus your reading on the topics covered in the lecture. Readings have to be done prior to class as corresponding quizzes will be delivered at the beginning of class.
4. Your note taking is essential. Pay attention to what we talk in class – this is one way to draw your attention to important concepts.
5. Please review your notes as well as the ppt slides at least once for 15 minutes within 12 hours after each lecture. This 15-minute investment can yield large dividends.

6. The rule of thumb is 2 hours of study outside of class for every hour spent in class per week (Our 3-credit hour class – 6 hours of outside study time per week as a minimum). This is approximate and will depend on your level of preparation upon entering the course.
7. **Please attend office hours (at least once).** I would like to get to know each of you, and I want to ensure any questions you have are answered. If you can't make the indicated times, we can set an alternative time to meet.

Services and Student Support

Resources

Successful people access support when needed. Here are some free resources available to you:

- [The Learning Center](#) provides one-on-one academic coaching, peer tutoring, study workshops, and online tools to help students learn more efficiently and perform better in their coursework.
- [The Writing Center](#) provides online and in-person consultations and online resources to help students become better writers, as well as additional workshops and speaking/writing groups for ESL students.
- [The Undergraduate Testing Center](#) provides a secured, proctored environment for students who are unable to take exams at the normally scheduled time (pre-arranged).
- [CAPS \(Counseling and Psychological Services\)](#) provides consultation and connection to clinically appropriate services for short or long-term needs. Walk-in evaluations are available at their facilities on the third floor of the Campus Health Services building.
- [Academic Advising Program](#) assists students in their growth and development by discussing educational and career plans, making appropriate course selections, etc.
- [Office of the Dean of Students](#) assists students in successfully transitioning through a variety of challenging circumstances and situations that can impact their lives, both inside and outside the classroom.
- [University Career Services](#) to support the mission of the University and Student Affairs by providing personalized and innovative career education to empower student success and transformation.
- [LGBTQ Center](#) works to foster a welcoming and inclusive environment for students of all sexual orientations, gender identities, and gender expression.

Title IX Resources

Any student who is impacted by discrimination, harassment, interpersonal (relationship) violence, sexual violence, sexual exploitation, or stalking is encouraged to seek resources on campus or in the community. Please contact the Director of Title IX Compliance (Adrienne Allison – Adrienne.allison@unc.edu), Report and Response Coordinators in the Equal Opportunity and Compliance Office (reportandresponse@unc.edu), Counseling and Psychological Services (confidential), or the Gender Violence Services Coordinators (gvsc@unc.edu; confidential) to discuss your needs. Additional resources: safe.unc.edu.

Accessibility Resources and Services

Equal Opportunity and Compliance Accommodations Team ([Accommodations - UNC Equal Opportunity and Compliance](#)) receives requests for accommodations for disability, pregnancy and related conditions, and sincerely held religious beliefs and practices through the University's Policy on Accommodations. EOC Accommodations.

Academic Policies

Attendance Policy

As stated in the University's [Class Attendance Policy](#), no right or privilege exists that permits a student to be absent from any class meetings, except for these University Approved Absences:

1. Authorized University activities: [University Approved Absence Office \(UAAO\)](#) website provides information and [FAQs for students](#) and [FAQs for faculty](#) related to University Approved Absences
2. Disability/religious observance/pregnancy, as required by law and approved by the [Equal Opportunity and Compliance Office](#) (EOC)
3. Significant health condition and/or personal/family emergency as approved by the [Office of the Dean of Students](#), [Gender Violence Service Coordinators](#), and/or the [Equal Opportunity and Compliance Office](#) (EOC).

Class Policy

Please communicate with me early about potential absences. Please be aware that you are bound by the [Honor Code](#) when making a request for a University approved absence.

Course Schedule

*Tentative. An updated syllabus will be posted on Canvas if things change as the term progresses.

Date	Day	Lesson	Topic*	Kalat Chapter (14 th ed) & Assignments
Jan 09	Thurs		Introduction, Syllabus, Course Structure, "How to get involved in research at UNC"	
Jan 14	Tues	1	Neural Cells of the Nervous System I	1.2 & Paper on Canvas Quiz 1
Jan 16	Thurs	2	Neural Cells of the Nervous System II	
Jan 21	Tues	3	Neuronal Communication I: Action Potential	1.3 Quiz 2
Jan 23	Thurs	4	Neuronal Communication II: Synapse	2.1, 2.2 Quiz 3 Paper 1 due
Jan 28	Tues	5	Neuronal Communication III: Neurotransmitter	2.2 Quiz 4
Jan 30	Thurs	6	Structure of the Nervous System	3.1, 3.2 Quiz 5 Scientist Spotlight A#1
Feb 4	Tues		Exam 1	Lessons 1-6
Feb 6	Thurs	7	Research Methods	3.3 Quiz 6
Feb 11	Tues	8	Genetics & Evolution	4.1, 4.2, Quiz 7
Feb 13	Thurs	9	Visual System	5 Quiz 8 Scientist Spotlight A#2
Feb 18	Tues	10	Motor System	7 Quiz 9
Feb 20	Thurs	11	Sensory Systems: Auditory & Somatosensory	6.1 & 6.2 Quiz 10 Paper 2 due
Feb 25	Tues	12	Sensory Systems: Taste & Smell	6.3 Quiz 11
Feb 27	Thurs	13	Wakefulness & Sleep	8 Quiz 12 Scientist Spotlight A#3
Mar 4	Tues		Exam 2	Lessons 7-13
Mar 6	Thurs	14	Internal Regulation I	9 Quiz 13
Mar 11	Tues		Spring Break – No classes held	
Mar 13	Thurs		Spring Break – No classes held	
Mar 18	Tues	15	Internal Regulation II	9 Paper 3 due
Mar 20	Thurs	16	Learning and Memory I	12 Quiz 14
Mar 25	Tues	17	Learning and Memory II	12
Mar 27	Thurs	18	Emotional Behaviors I	11 Quiz 15
Apr 1	Tues	19	Emotional Behaviors II	11 Scientist Spotlight A#4
Apr 3	Thurs	20	Psychological Disorders I: Substance Use & Mood	14.1, 14.2 Quiz 16 Paper 4 due
Apr 8	Tues	21	Psychological Disorders I: Affective & Neurodevelopmental	14.3, 14.4
Apr 10	Thurs		Exam 3	Lessons 14-21
Apr 15	Tues	22	Scientist Spotlight Presentation	Groups 1-5
Apr 17	Thurs		Well-being Day – No classes held	
Apr 22	Tues	23	Scientist Spotlight Presentation	Groups 6-9
Apr 24	Thurs	24	LDOC – FINAL EXAM Review	Exams 1-3
May 1	Thurs		Final Exam From 12:00 PM to 3:00 PM	In-class